EG1311: Design & Make

Serial Communication

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Blink Demo

```
void setup() {
    pinMode(13, OUTPUT);
}

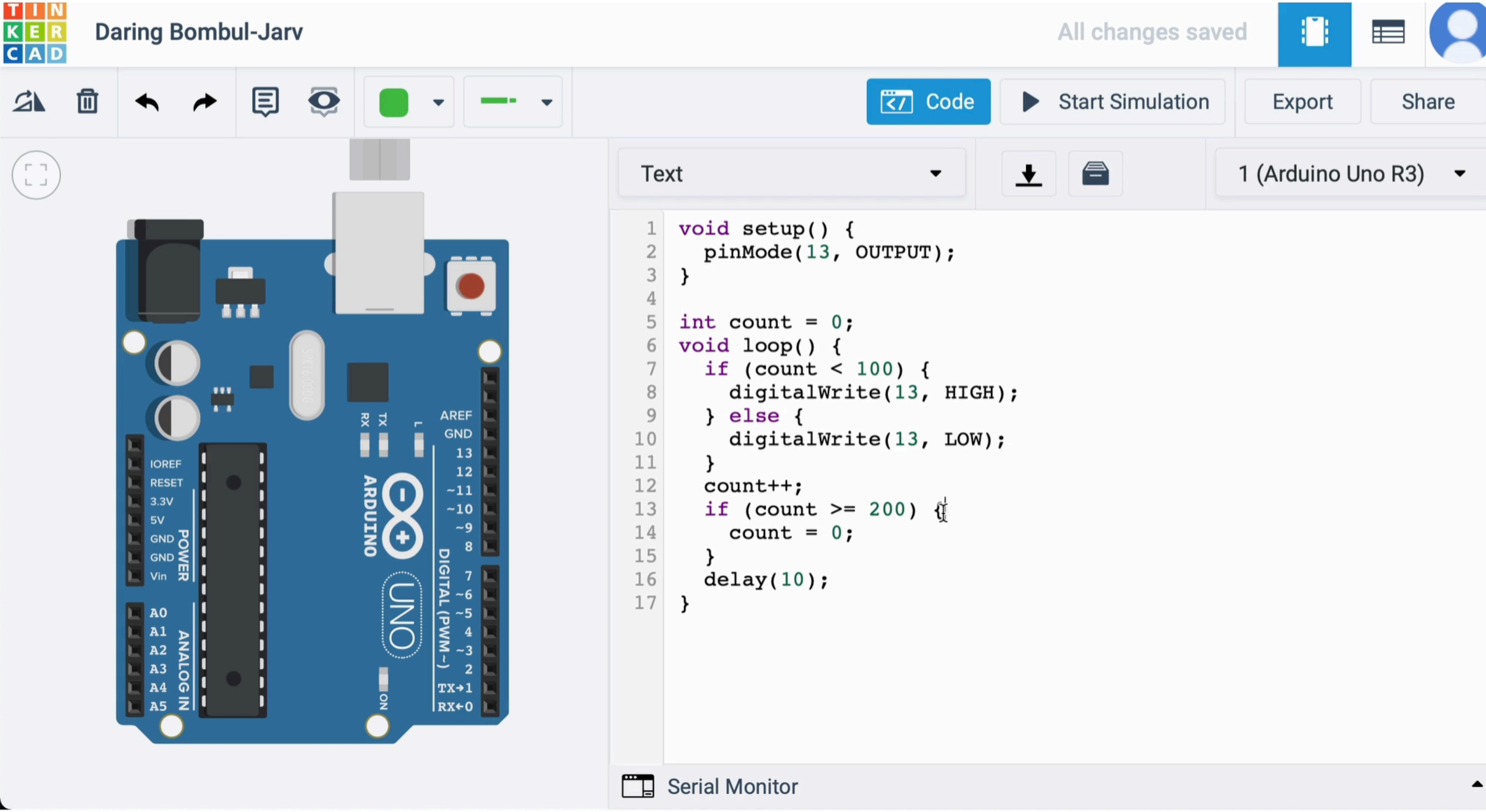
void loop() {
    digitalWrite(13, HIGH);
    delay(1000);
    digitalWrite(13, LOW);
    delay(1000);
}
```

Blink Demo

```
void setup() {
    pinMode(13, OUTPUT);
}

int count = 0;  // times through loop
void loop() {
    if (count < 100) {
        digitalWrite(13, HIGH);
    } else {
        digitalWrite(13, LOW);
    }
    count++;
    if (count >= 200) {
        count = 0;
    }
    delay(10);
}
```

Blink Demo

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void setup() {
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}

int count = 0;
void loop() {
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  }
  count++;
  if (count >= 200) {
    count = 0;
  }
  delay(10);
}
```

Serial Communication

- Protocol for computers to talk to each other
- Use two one-way wires TX->RX, RX->TX
- Sends data serially, byte-by-byte
- Baudrate for Arduino `9600` or `115200`

```
void Serial.begin(baudrate);
void Serial.print(string);
void Serial.println(string);
```

Serial Demo

```
void setup() {
    pinMode(13, OUTPUT);
    Serial.begin(9600);
}

int count = 0;  // times through loop
void loop() {
    Serial.println(count);
    if (count < 100) {
        digitalWrite(13, HIGH);
    } else {
        digitalWrite(13, LOW);
    }
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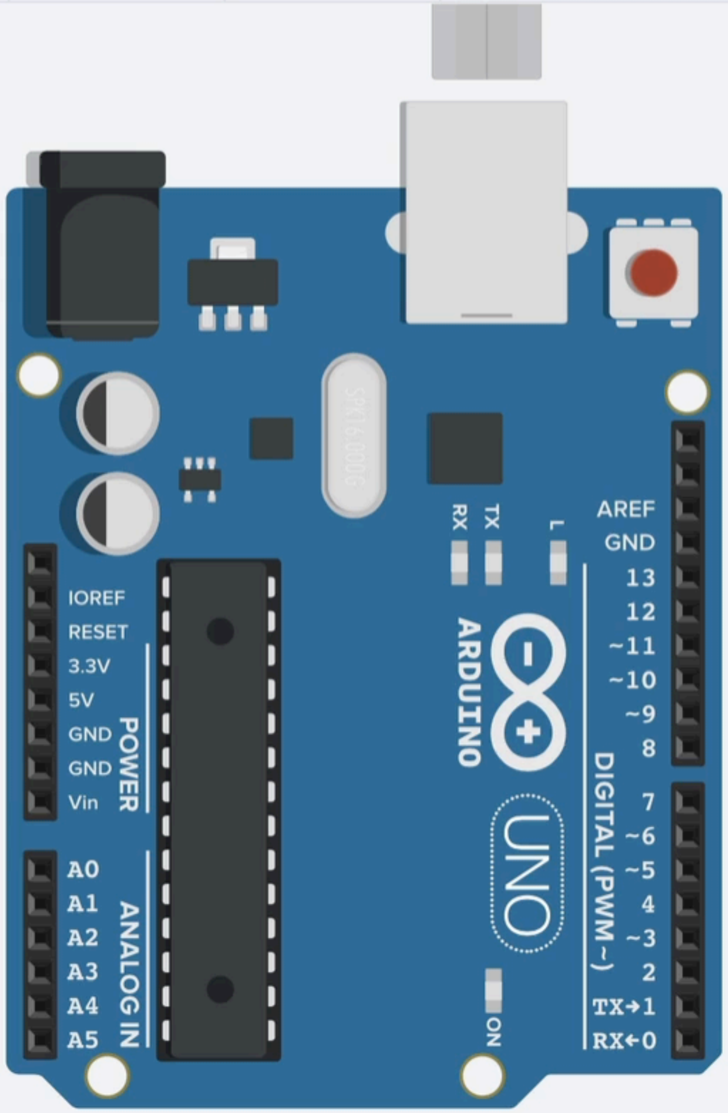
    if (count >= 200) {
        count = 0;
    }
    delay(10);
}
```

Serial Demo

Daring Bombul-Jarv

All changes saved

Code | Start Simulation | Export | Share



Text

1 (Arduino Uno R3)

```
void setup() {
  pinMode(13, OUTPUT);
  Serial.begin(9600);
}

int count = 0;
void loop() {
  Serial.println(count);
  if (count < 100) {
    digitalWrite(13, HIGH);
  } else {
    digitalWrite(13, LOW);
  }
  count++;
  if (count >= 200) {
    count = 0;
  }
  delay(10);
}
```

Serial Monitor

Serial Demo

Arduino File Edit Sketch Tools Help

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Serial Monitor

blink

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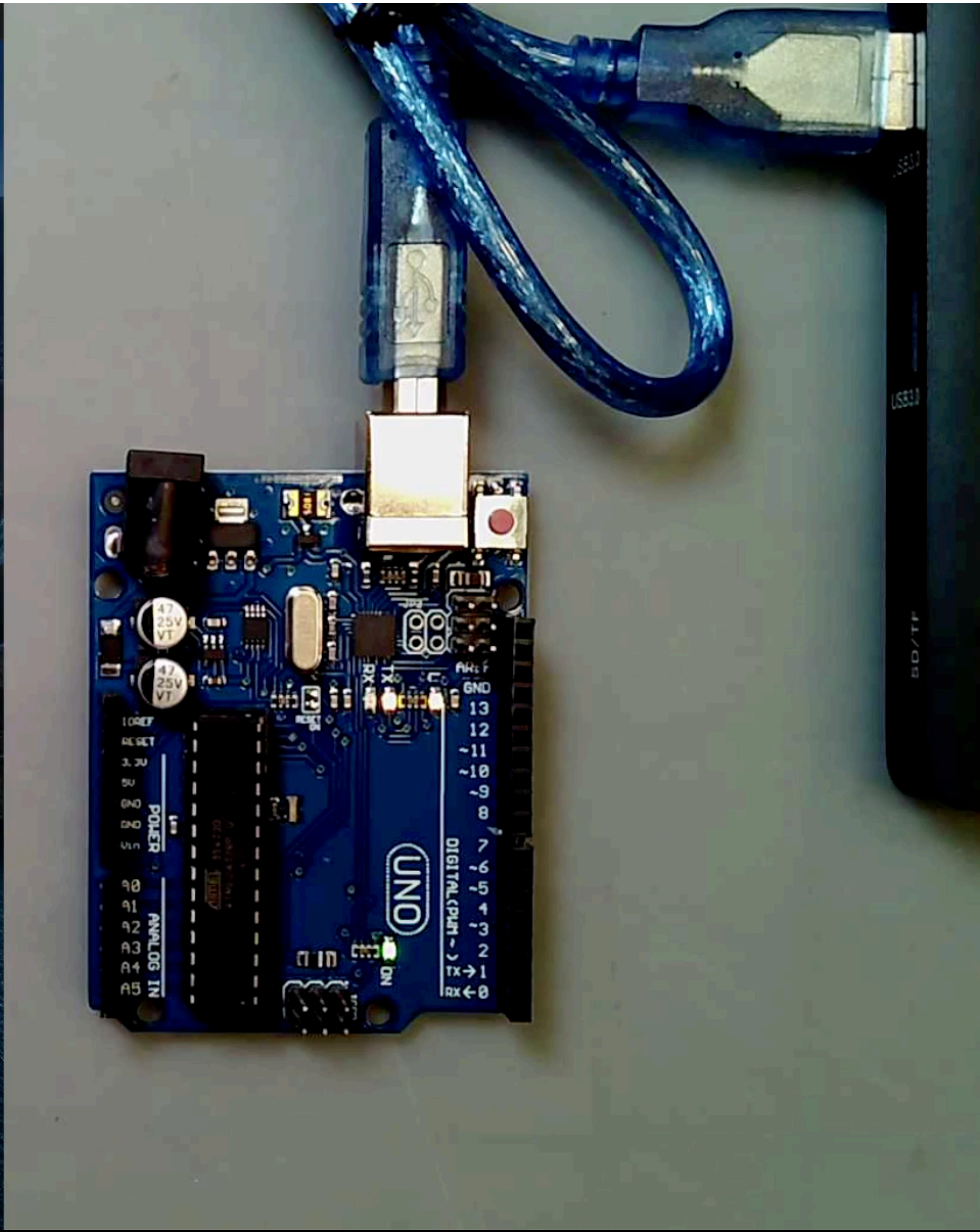
    count++;
    if (count >= 200) {
        count = 0;
    }
    delay(10);
}
```

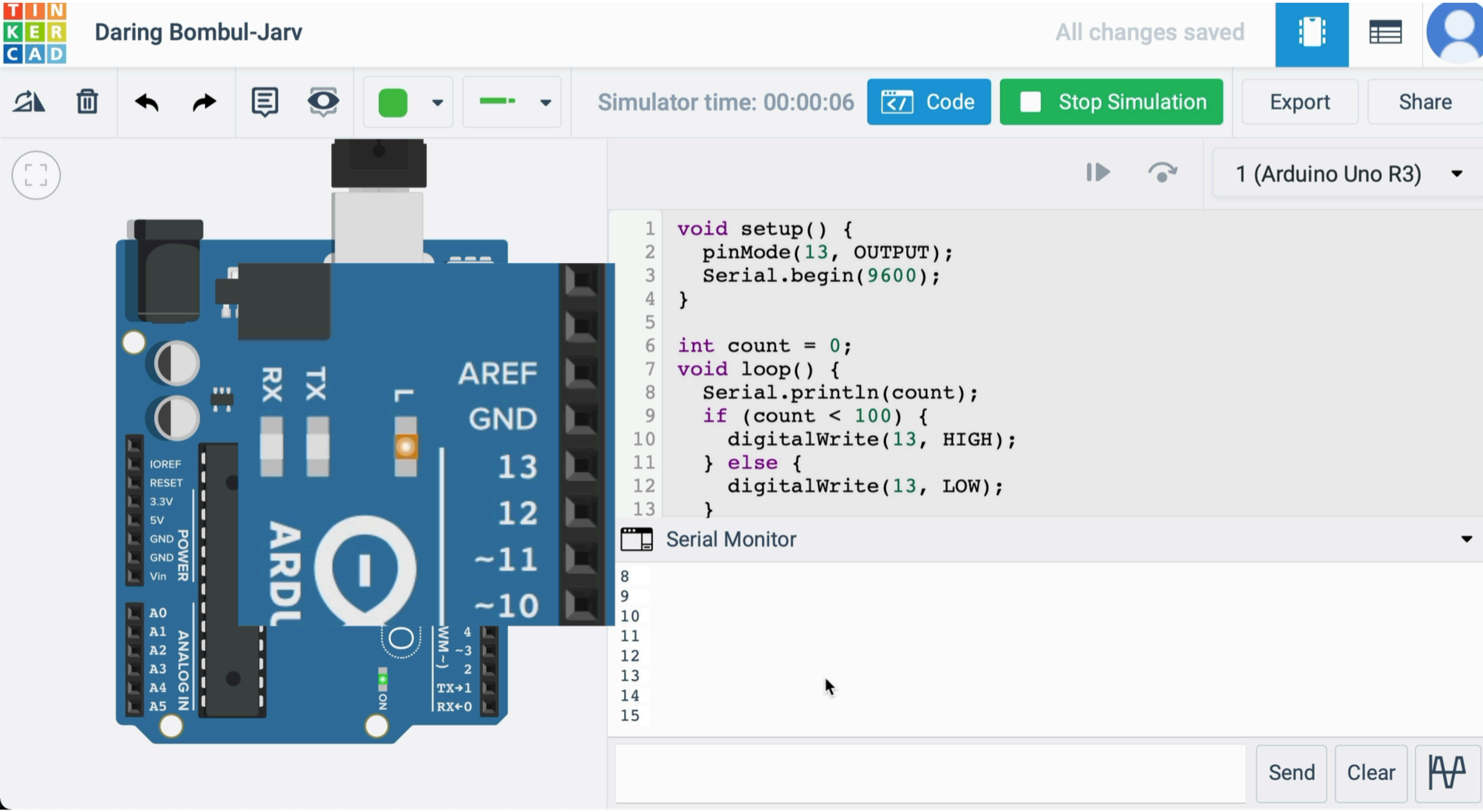
Done uploading.

```
Sketch uses 2200 bytes (6%) of program storage space. Max
Global variables use 190 bytes (9%) of dynamic memory, le
```

1 Arduino Uno on /dev/cu.usbmodem142301



Serial Demo



Daring Bombul-Jarv
All changes saved
Simulator time: 00:00:06
Code
Stop Simulation
Export
Share
1 (Arduino Uno R3)
void setup() {
pinMode(13, OUTPUT);
Serial.begin(9600);
}
int count = 0;
void loop() {
Serial.println(count);
if (count < 100) {
digitalWrite(13, HIGH);
} else {
digitalWrite(13, LOW);
}
Serial Monitor
8
9
10
11
12
13
14
15
Send
Clear
AREF
GND
13
12
~11
~10
RX
TX
L
IOREF
RESET
3.3V
5V
GND
GND
Vin
POWER
A0
A1
A2
A3
A4
A5
ANALOG IN
ON
TX→1
RX←0

Timer Demo

```
void setup() {
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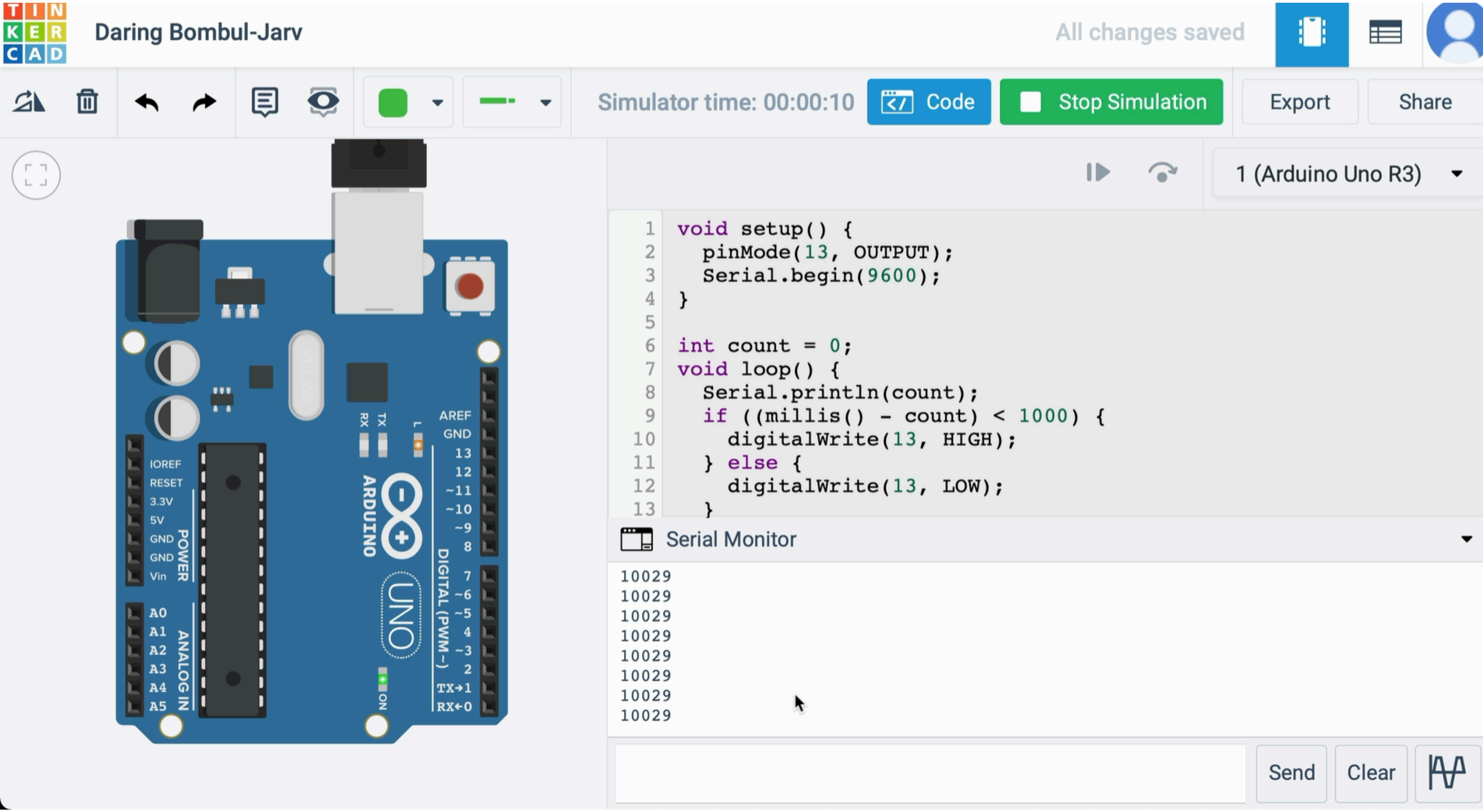
}
```

Timer Demo

```
void setup() {
    pinMode(13, OUTPUT);
    Serial.begin(9600);
}

int count = 0;  // time since last change
void loop() {
    Serial.println(count);
    if ((millis() - count) < 1000) {
        digitalWrite(13, HIGH);
    } else {
        digitalWrite(13, LOW);
    }
    if ((millis() - count) >= 2000) {
        count = millis();
    }
    delay(10);
}
```

Timer Demo


Daring Bombul-Jarv
All changes saved
Simulator time: 00:00:10
Code
Stop Simulation
Export
Share
1 (Arduino Uno R3)
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  } else {
    digitalWrite(13, LOW);
  }
Serial Monitor
10029
10029
10029
10029
10029
10029
10029
10029
Send
Clear
ARDUINO
UNO
DIGITAL (PWM~)
ANALOG IN
POWER

Timer Demo

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Upload

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void setup() {
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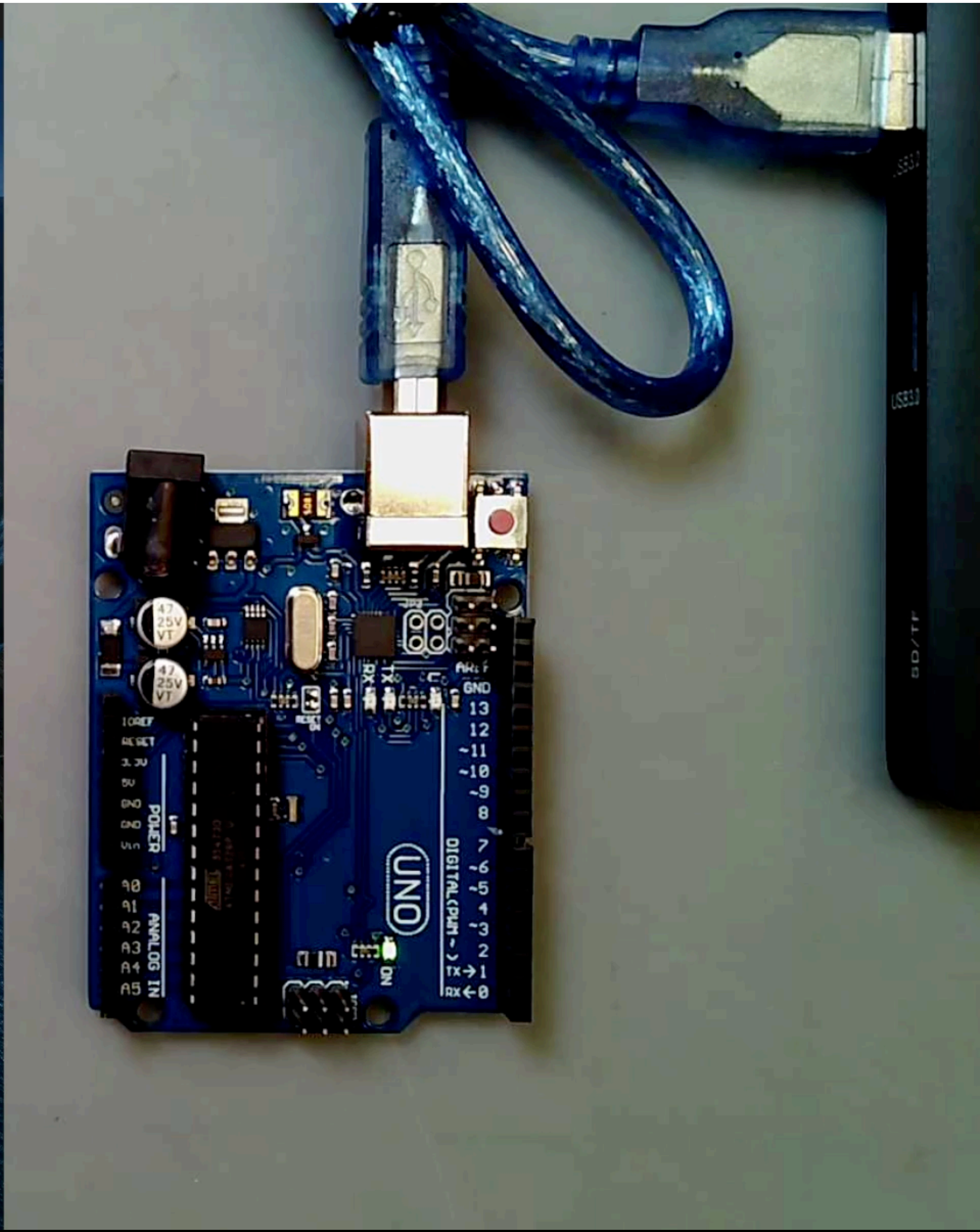
18 Arduino Uno on /dev/cu.usbmodem142301



Timer Demo

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void setup() {
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}

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void loop() {
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    if ((millis() - count) < 1000) {
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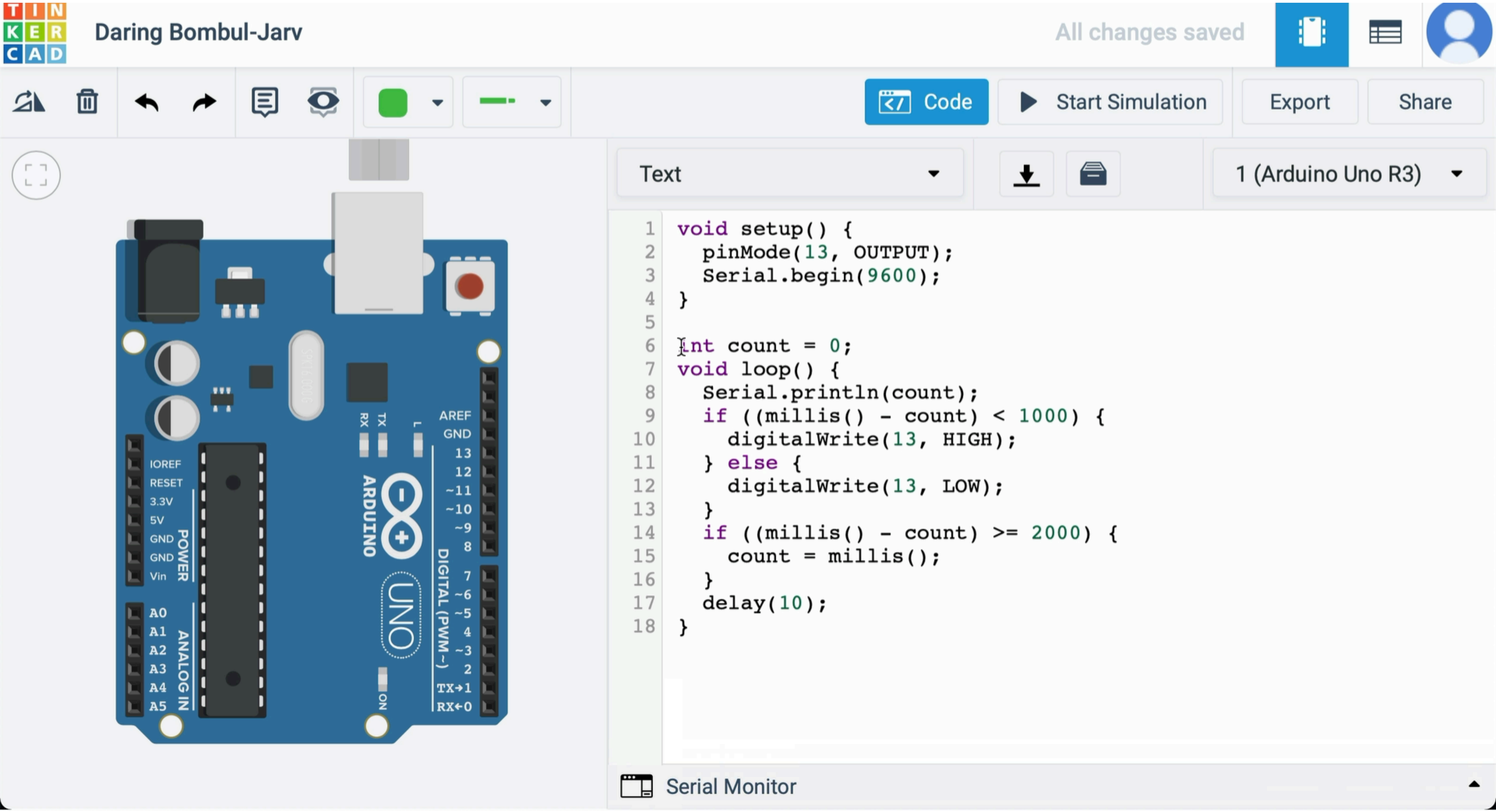
    }
    if ((millis() - count) >= 2000) {
        count = millis();
    }
    delay(10);
}
```

Timer Demo

```
void setup() {
    pinMode(13, OUTPUT);
    Serial.begin(9600);
}

unsigned long count = 0;
void loop() {
    Serial.println(count);
    if ((millis() - count) < 1000) {
        digitalWrite(13, HIGH);
    } else {
        digitalWrite(13, LOW);
    }
    if ((millis() - count) >= 2000) {
        count = millis();
    }
    delay(10);
}
```

Timer Demo


TINKERCAD
Daring Bombul-Jarv
All changes saved
Code
Start Simulation
Export
Share
Text
1 (Arduino Uno R3)
void setup() {
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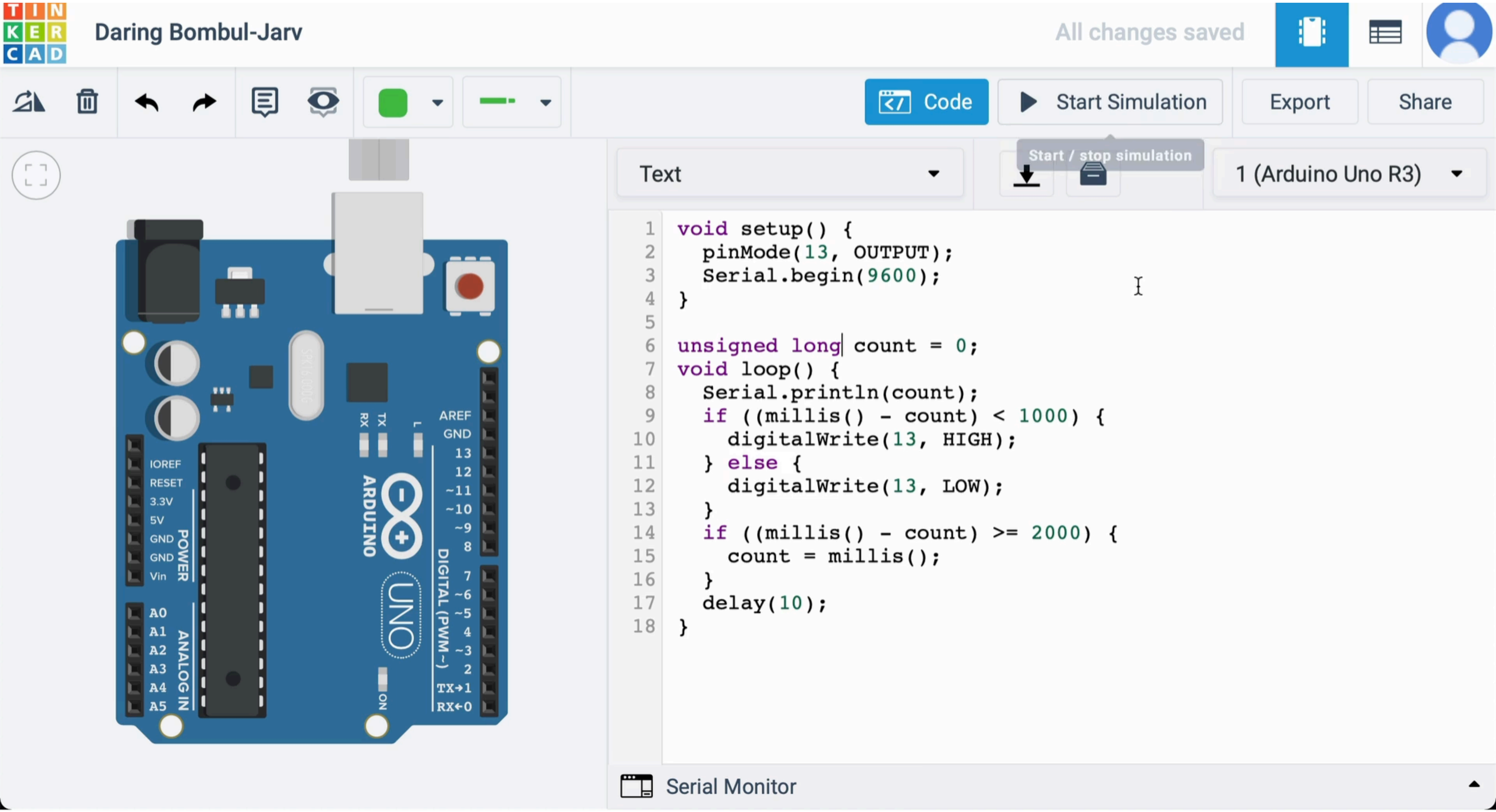
  }
  delay(10);
}
Serial Monitor
AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX→1
RX←0
DIGITAL (PWM ~)
ARDUINO
UNO
IOREF
RESET
3.3V
5V
GND
GND
Vin
POWER
A0
A1
A2
A3
A4
A5
ANALOG IN
RX
TX
L
ON

Timer Demo


Daring Bombul-Jarv
All changes saved
Code
Start Simulation
Export
Share
Start / stop simulation
Text
1 (Arduino Uno R3)
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ANALOG IN

Timer Demo

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```

```
Sketch uses 2296 bytes (7%) of program storage space. Max
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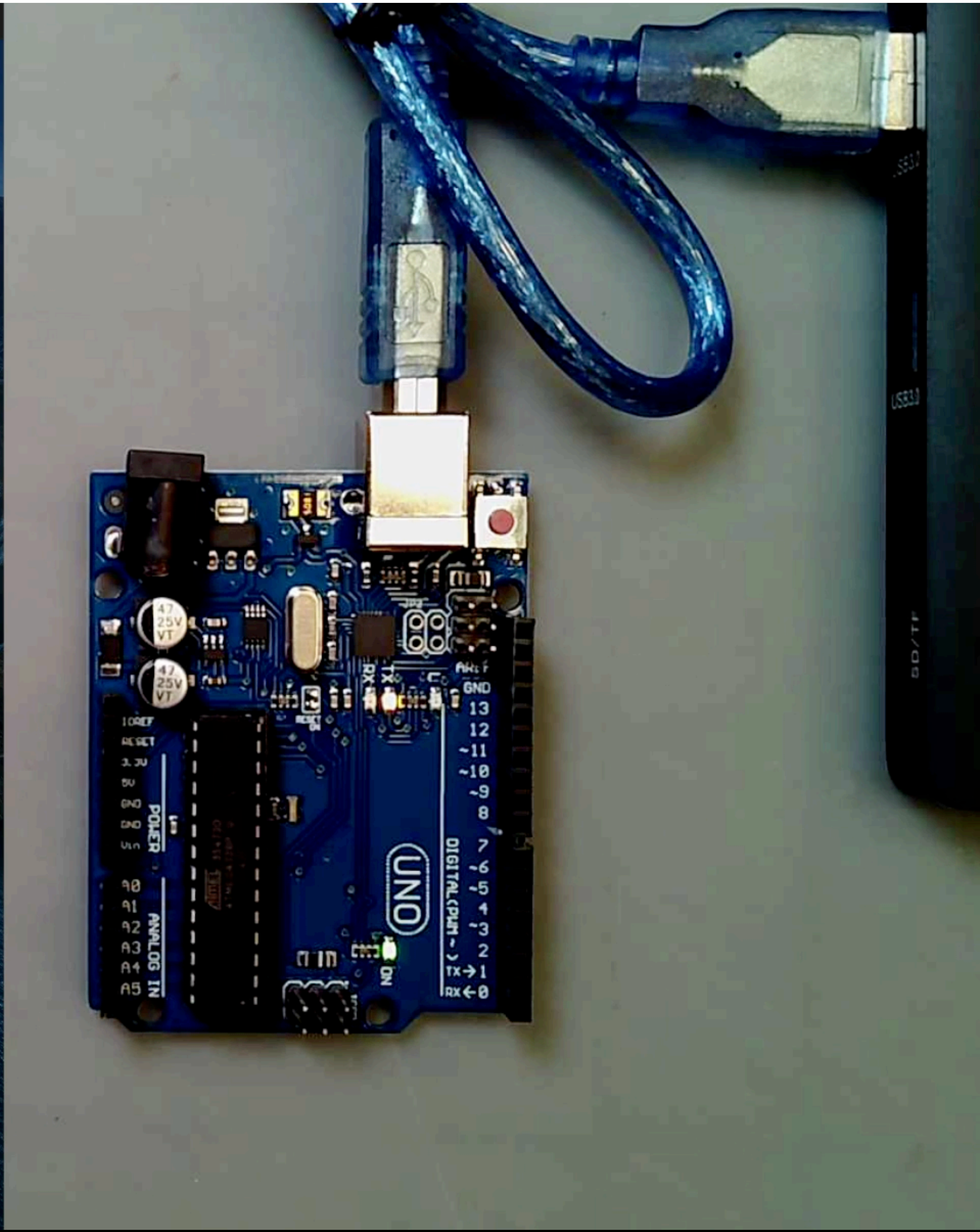
6 Arduino Uno on /dev/cu.usbmodem142301



Timer Demo

Arduino File Edit Sketch Tools Help

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Upload

blink §

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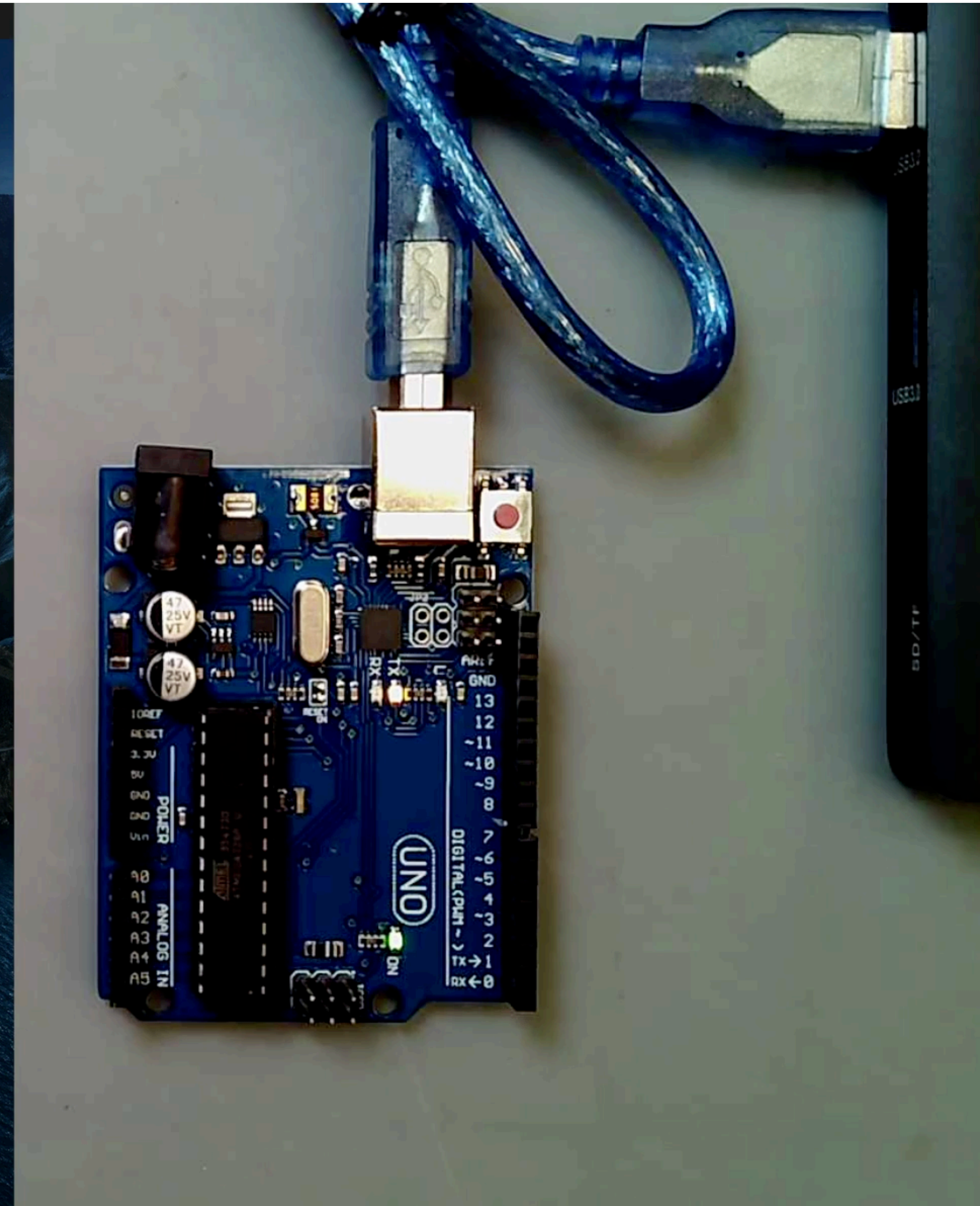
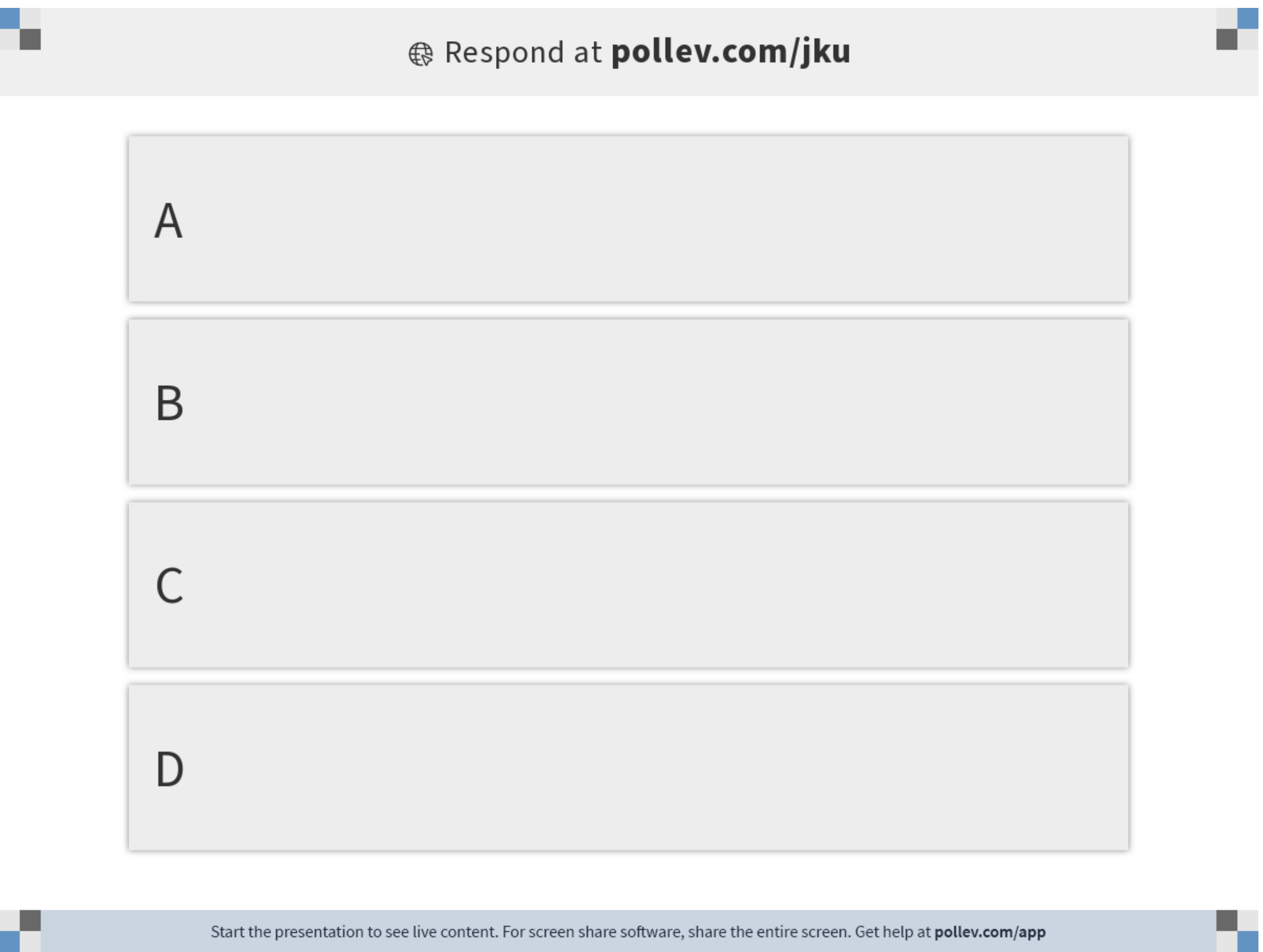


Practice

What is the default baudrate of the Arduino UNO Serial port?



A) 800 bytes/sec

B) 1000 bytes/sec

C) 1200 bytes/sec

D) 1400 bytes/sec

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